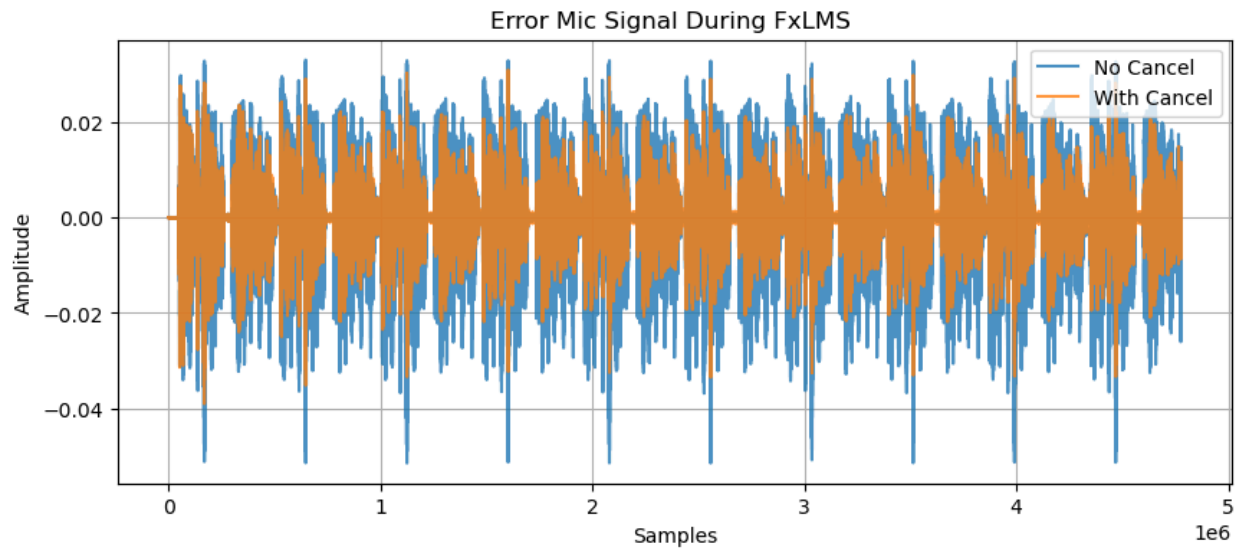
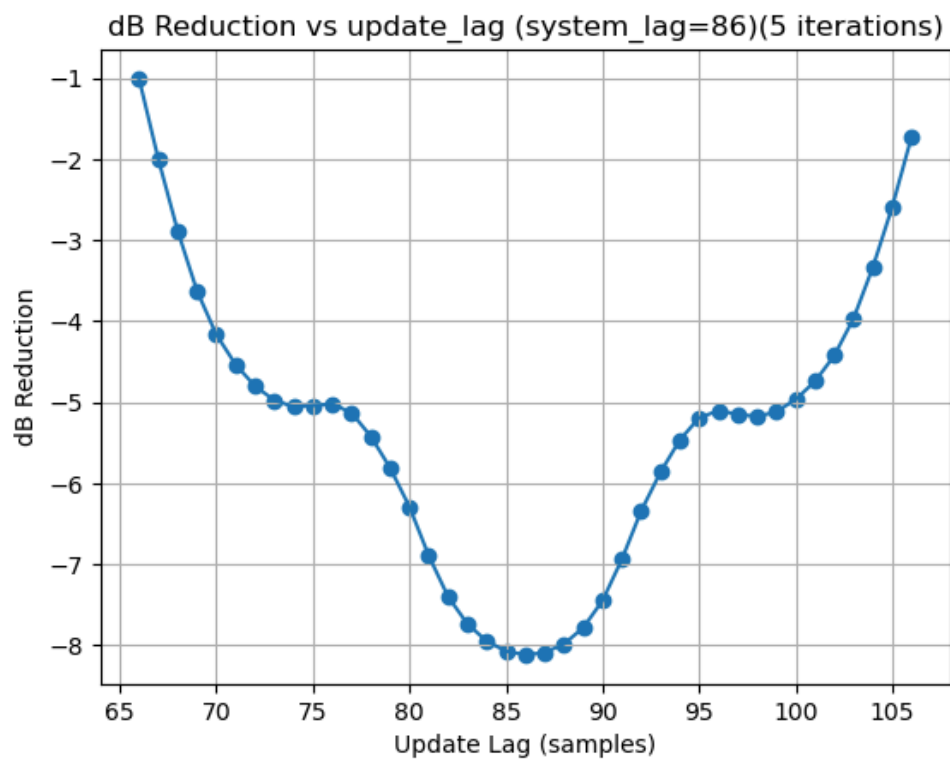


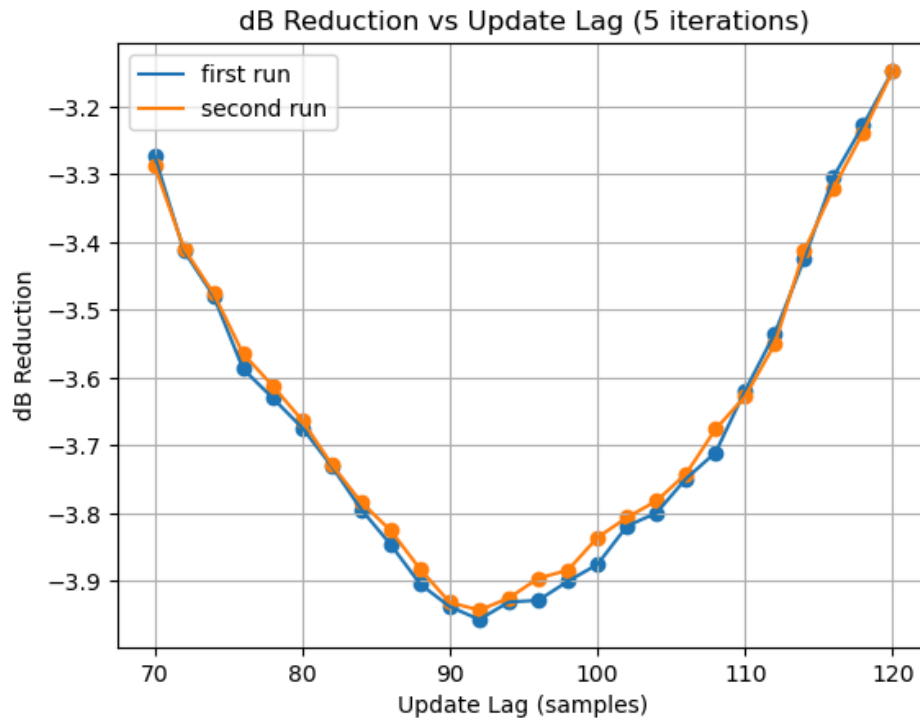
Fixing the update lag bug improved by like 1.5-2dB

- -5.83dB



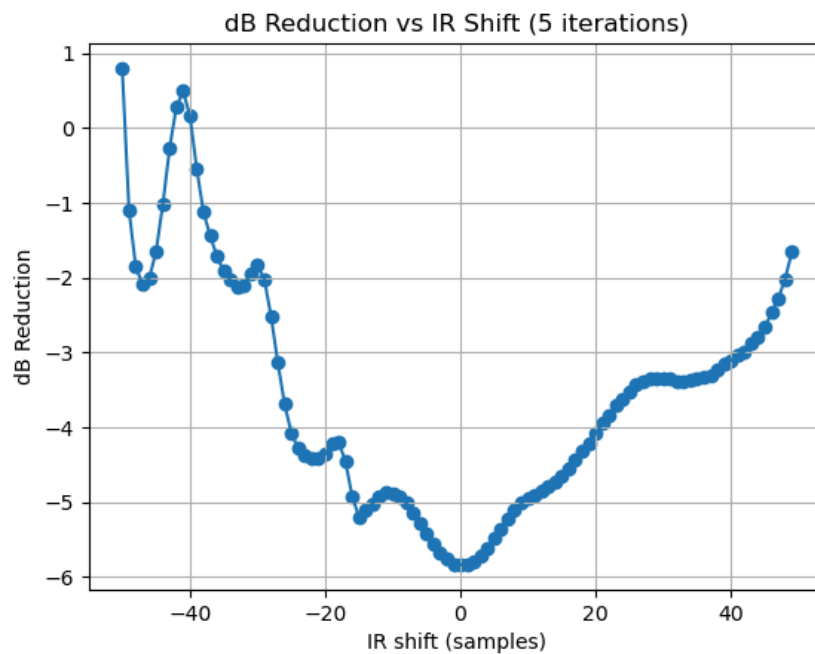
Update lag vs system lag misalignment





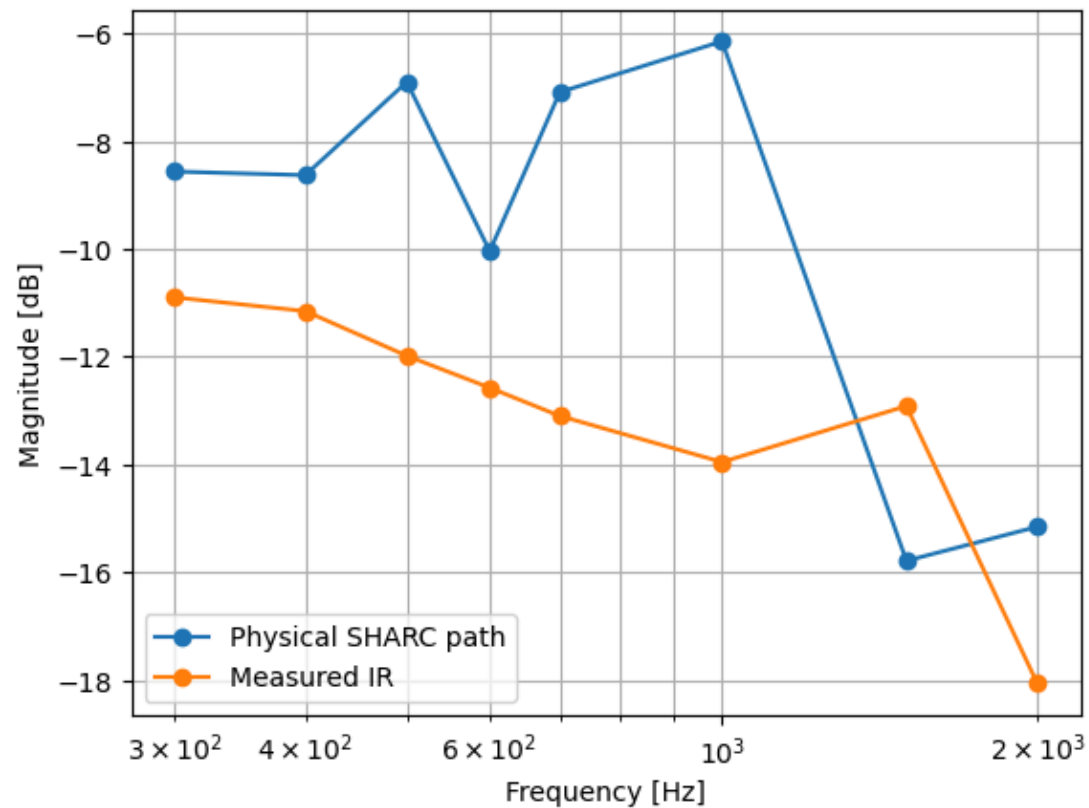
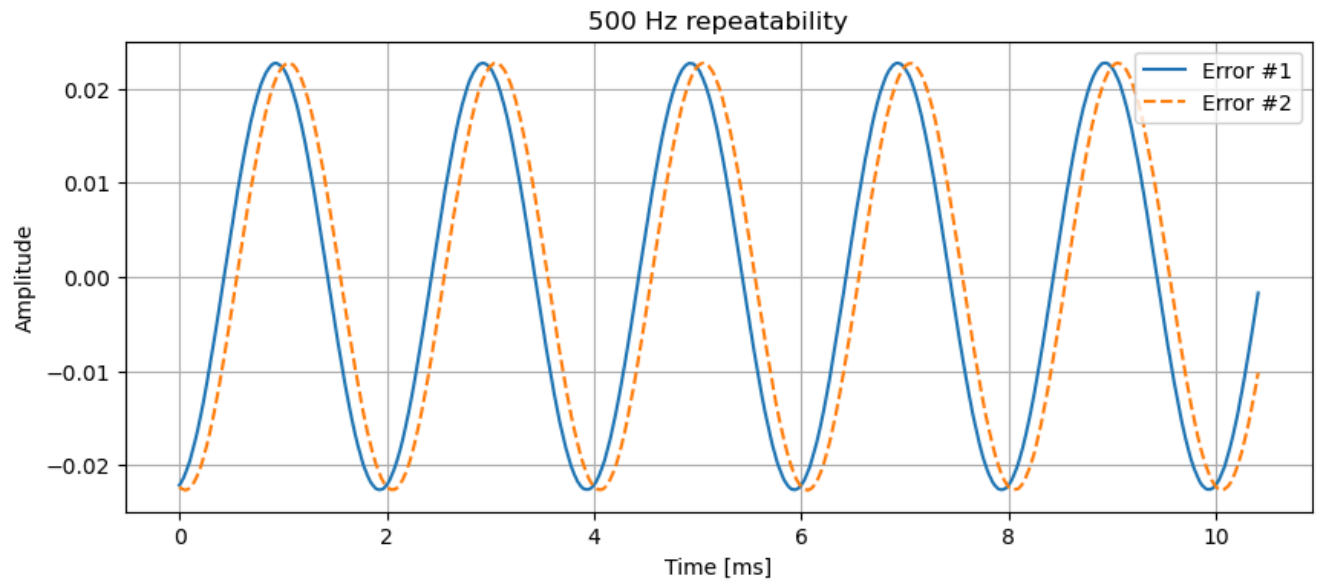
Performance vs IR shift

- Fixed physical model IR, shifted IR for Fx algorithm
- The controller is very sensitive to phase change
- Perhaps our alignment method is incorrect?



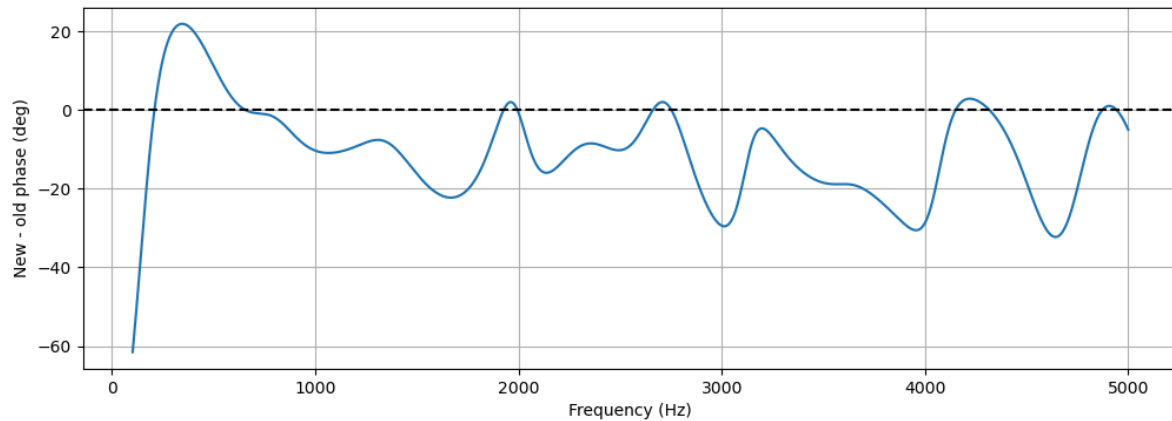
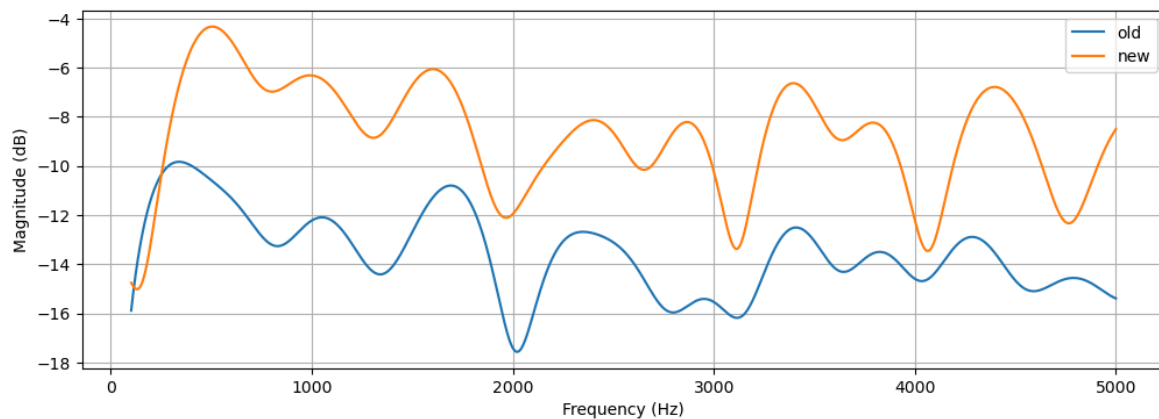
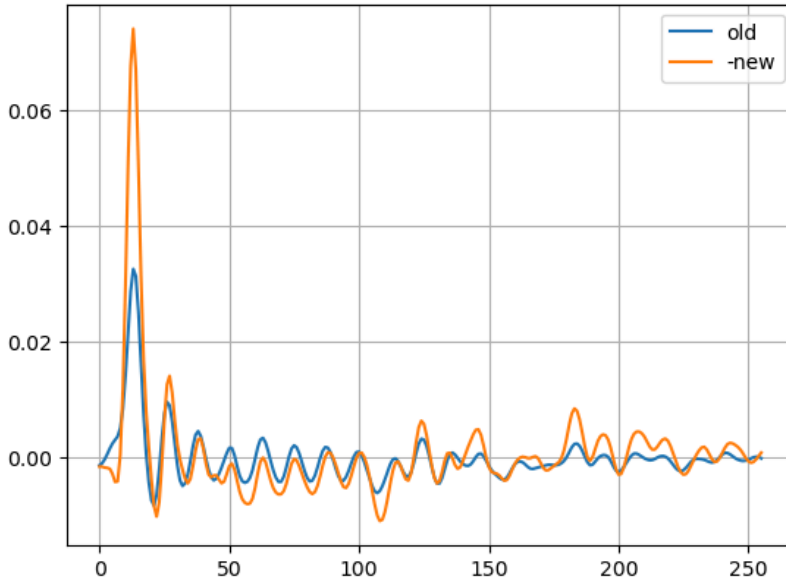
Different frequency tones measured through the Sharc vs IR

- Played a tones at different frequencies through the sharc
- Projected them onto sin/cosines to calculate the amplitude
- Compared it to the DTFT of the panel IR



IR measured through the Sharc

- Played a sine sweep through the Sharc on the panel instead of the Scarlett
- Different peak amplitude and some phase shift
- Polarity was reversed, had to negate it

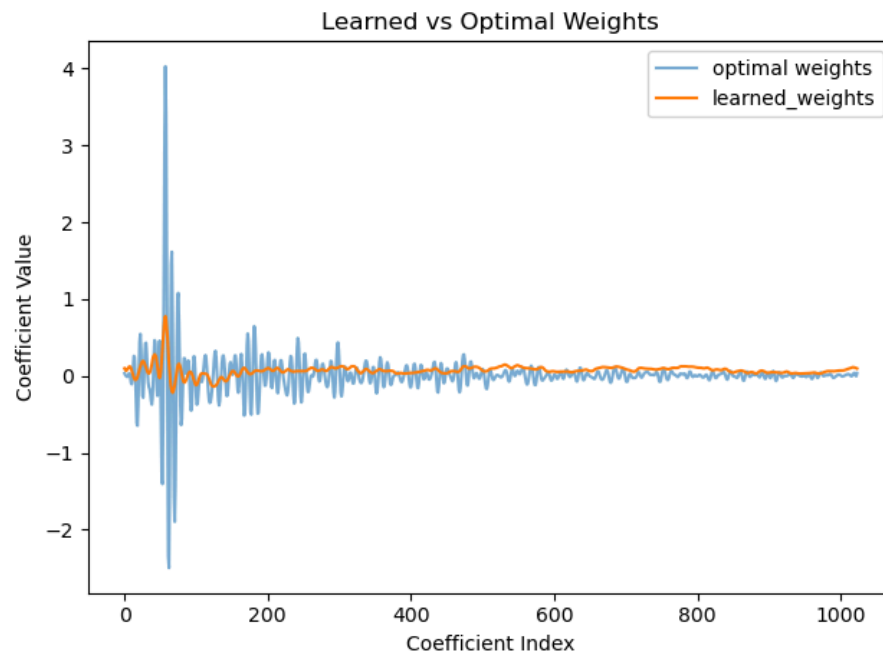
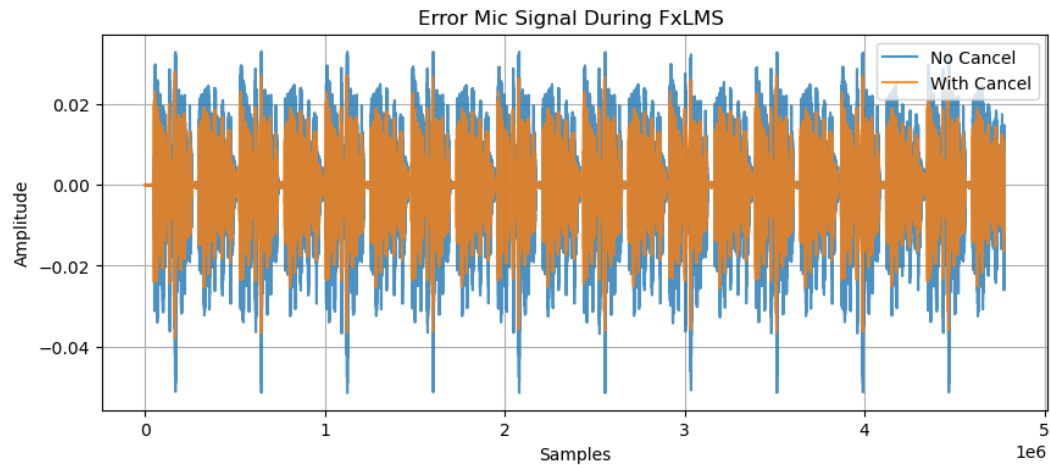


Learning comparison of IR combinations

Update IR	Physical Model IR	dB
Scarlett	Scarlett	-10.23 dB
Scarlett	Negated SHARC	-11.76 dB
Negated SHARC	Negated SHARC	-10.72 dB
Negated SHARC	Scarlett	-9.36 dB

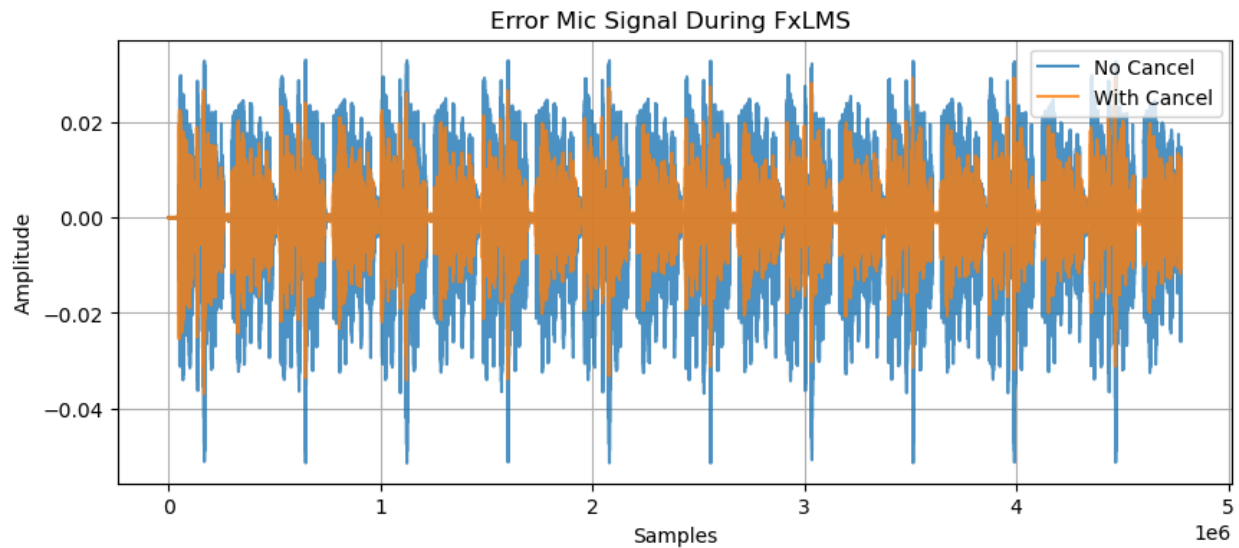
The lagged optimal filter kind of works online?

- -3.39 dB
- Different gain settings (0.15 offline, 0.1 real time)



Learning from the optimal filter did better

- -6.2dB



- Some of the improvement is less learning to do at the start
- The weight norm trends toward the from scratch norm

